

Econ 131
Spring 2024

Midterm **Solutions**

March 2024

Student Name:

Student ID:

Exam Instructions

- **Explanation should be written using pens (we recommend black or blue ink, as these often scan the best).** No pencils, except for graphs.
- **Do not write your solutions on pages that say “Do not write on this page”.** Answers written on these pages will not be graded.
- **Show your work.** Credit will only be awarded on the basis of what is written on the exam.
- **Affirm the academic honesty pledge below.** For those writing on a non-printed copy, please just write “Academic Honesty Pledge as on exam”, and sign your name.
If you do not affirm this pledge, your exam will be marked invalid.

0. ACADEMIC HONESTY PLEDGE

I confirm that I have abided by all academic honesty rules for UC Berkeley and Economics 131. I confirm that I did not see this exam before my official exam start time. I confirm that I have not shared and will not share this exam with anyone else.

Signature: _____

1. True/False/Uncertain Questions (12 points total)

- (a) A reduction in the marginal income tax rate should always make people work more.

FALSE. Depends on the relative magnitude of the income and substitution effects. If the income effect dominates a reduction in the marginal income tax will make people work *less*.

- (b) Once you have established that a treatment group and control group exhibit parallel trends in the period before a tax change, it is valid to estimate the effect of the tax change using the difference-in-differences estimator we studied in class:

$$DD = [Y^{T,After} - Y^{C,After}] - [Y^{T,Before} - Y^{C,Before}]$$

TRUE. Parallel pre-intervention trends (also called “pre-period trends”) are the key assumption needed for the DD estimator to be valid. The reason is that the DD estimator is valid when – in the absence of the intervention – the treatment group’s outcomes would have trended similarly to the control group’s outcomes. Parallel pre-trends are the key way to validate that assumption.

- (c) The U.S. poverty rate stayed constant over the 1972-2005 period discussed in class.

UNCERTAIN (THOUGH COULD GIVE ANY ANSWER AS LONG AS YOU DEFEND IT). The official poverty rate stayed roughly constant. However, the official poverty rate uses an inflation series that overstates inflation and also excludes tax credits like the EITC. After using the better inflation series (“CPI-RS” which better accounts for substitution bias and other biases) and tax credits like the EITC, the U.S. poverty declined substantially 1972-2005.

- (d) Suppose that the supply curve for widgets (in a standard graph with price on the y-axis and quantity on the x-axis) has a slope of 1, while the demand curve for widgets has a slope that is -1000. Most of the incidence of a tax on widgets will be on consumers.

TRUE. The less elastic side of the market bears a greater share of the tax burden. This is a case of relatively inelastic demand: consumer demand is less sensitive to price changes than supply is.

- (e) If earnings do not respond to taxes, then a utilitarian government would want to set taxes and transfers in order to equalize everyone’s consumption.

TRUE. As long as the marginal utility of consumption declines with consumption, the sum of utilities across households for a given level of aggregate consumption is maximized when every household consumes the same amount. If earnings do not respond to taxes, then there is no efficiency cost to achieving the equity gains of redistribution, so the government will set taxes and transfer in order to equalize everyone’s consumption. For example, that could be achieved with a 100% tax rate on earnings and using that tax revenue to fund an equal transfer of dollars to each household.

- (f) The experience of Denmark indicates that high earners do not decide where to live based on taxes.

FALSE. The 1991 Danish tax reform provided a tax break for high-earning foreigners choosing to reside in Denmark. In response, the number of high earning foreigners in Denmark rose dramatically, relative to slightly lower earning foreigners.

2. Income Taxation and Labor Supply 16 points total)

Assume that households have the following utility function over *consumption* and *labor*, given by:

$$U(c, l) = c - \frac{l^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}$$

where c denotes consumption, l denotes hours of labor supply, and $\eta > 0$ is nonzero. Households' only source of earnings is labor income. The hourly wage is w and the price of consumption is 1. The government collects revenue through a "flat tax" at rate t on earnings z : a household with pre-tax earnings z pays tz in taxes. The government also sends households a lump-sum transfer $T > 0$.

- (a) What is the household budget constraint with this tax and transfer system? How does this lump-sum transfer differ from the actual implementation of transfers in the United States? (3 points)

The household budget constraint with this tax and transfer system is $c = (1-t)wl + T$. (2 points)

The lump-sum transfer T differs from the actual implementation of transfers in the United States. For instance the Earned Income Tax Credit (EITC) does not send households a fixed lump-sum but rather a credit that depends on their earnings (i.e. $T(wl)$). Additionally, the transfers sent to households are contingent on the number of children (i.e. $T(\text{number of children})$). (1 points)

- (b) Solve for the household's optimal consumption c^* and labor supply choice l^* as functions of w, η, t and T . (4 points)

First, we must write down the optimization problem. We could either write the problem as a a *Lagrangian*:

$$\max_{c, l, \lambda} L = c - \frac{l^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}} - \lambda[c - (1-t)wl - T], \quad (1)$$

or *directly substitute* the budget $c(l) = (1-t)wl + T$ into the utility function $U(c, l)$ to generate a budget-inclusive utility function $\tilde{U}(l)$:

$$\max_l \tilde{U}(l) = U(c(l), l) = [(1-t)wl + T] - \frac{l^{1+\frac{1}{\eta}}}{1+\frac{1}{\eta}}. \quad (2)$$

Either method will yield the same solution. (2 points)

Second, we must solve for the optimal labor supply l^* . Since the second method only requires us to write down one first-order condition instead of three I will continue along that route. The first-order condition for labor supply is:

$$\frac{d\tilde{U}}{dl} : (1-t)w - l^{\frac{1}{\eta}} = 0. \quad (3)$$

Rearranging we find that the optimal labor supply choice of households is $l^* = [(1 - t)w]^\eta$. (1 points)

Finally, we must solve for the optimal consumption choice c^* . Luckily with either the *Lagrangian* or *direct-substitution* method we can use the budget constraint to solve for the optimal consumption choice:

$$c^* = (1 - t)wl^* + T = (1 - t)w[(1 - t)w]^\eta + T = [(1 - t)w]^{1+\eta} + T. (1points) \quad (4)$$

- (c) Compute the taxable income elasticity (i.e. the elasticity of labor supply with respect to post-tax wage). Based on empirical findings discussed in class, is this value usually greater than or less than 1 for top income earners? (3 points)

Since the post-tax wage is $(1 - t)w$ and the labor supply choice is $l^* = [(1 - t)w]^\eta$, the taxable income elasticity (TIE) is:

$$TIE = \frac{dl^*}{d[(1 - t)w]} \frac{[(1 - t)w]}{l^*} = (\eta[(1 - t)w]^{\eta-1}) \left(\frac{[(1 - t)w]}{[(1 - t)w]^\eta} \right) = \eta (2points) \quad (5)$$

Based on empirical findings discussed in class this elasticity is around .1-.5 (see lecture 8, slides 25 - 39). However, there is considerable uncertainty about this parameter. (1 points)

- (d) What is the elasticity of the labor supply with respect to the lump-sum transfer? Given this elasticity, does the income effect dominate the substitution effect? (3 point)

Since the labor supply choice is $l^* = [(1-t)w]^\eta$ is not a function of the transfer T the elasticity is zero. (1 points)

Lump-sum transfers T have a direct income effect on households but no substitution effect – since they do not affect the post-tax wage $(1-t)w$. Because the lump-sum transfer T has no effect on the labor supply choice it's clear that income effects DO NOT dominate substitution effects. Alternatively, we could look back at the taxable income elasticity and note that $TIE = \eta > 0$. Therefore raising taxes would decrease labor supply, again implying that income effects DO NOT dominate substitution effects. (2 points)

- (e) Let $l^d(w) = w^{-\eta^d}$ be the labor demand of firms. What will be the equilibrium wage w^* as a function of η, η^d , and t ? (3 points)

In order to solve the equilibrium wage w^* we set labor supply l^* equal to labor demand l^d :

$$[(1-t)w]^\eta = w^{-\eta^d}. (1points) \tag{6}$$

Next we rearrange this equation to solve for the equilibrium wage $w^* = [(1-t)]^{-\frac{\eta}{\eta+\eta^d}}$. (2 points)

3. Government Budget 16 points total)

Assume the government's only source of revenue each year t is income taxation and its only expenditure is defense spending, denoted by G_t . Let τ_t be the linear tax rate on income in year t , and let $z(\tau) = (1 - \tau)^\phi$ be the income level for a given tax rate. Let the growth rate of the economy be zero $g = 0$.

- (a) For a given tax rate τ_t , what is the tax revenue of the government as a function of τ_t and ϕ ? (2 point)

For a given tax rate τ_t the pre-tax income level of the economy will be $z(\tau_t) = (1 - \tau_t)^\phi$. Because the income tax is applied linearly the tax revenue (TR) will be:

$$TR(\tau_t) = \tau_t z(\tau_t) = \tau_t (1 - \tau_t)^\phi. (2point) \quad (7)$$

- (b) What is the primary deficit each period as a function of τ_t, ϕ , and G_t ? (2 point)

The primary deficit is the difference between government expenses and revenues. The only expenditure of the government is defense spending G_t and the only source of revenue is from the income tax $TR(\tau_t)$. Therefore the primary deficit is $G_t - TR(\tau_t)$ which equals $G_t - \tau_t(1 - \tau_t)^\phi$ (2 point)

- (c) If the government runs a balanced budget (i.e., defense spending equals tax revenue) every year, what is the maximum spending on defense G_t that the government can achieve as a function ϕ ? (4 point)

If the government runs a balanced budget the primary deficit is zero and defense spending equals income tax revenues $G_t = \tau_t(1 - \tau_t)^\phi$. (1 point)

The government's ability to pay for defense spending G_t is only limited by their ability to raise revenue $TR(\tau_t)$. Because income declines as the tax rate increases the government faces a Laffer curve. The tax rate that maximizes income tax revenue $\tau_t^* = \arg \max \tau_t(1 - \tau_t)^\phi$ is the tax rate at the peak of the Laffer curve. We could either solve this optimization problem or remember from class that $\tau_t^* = \frac{1}{1+\phi}$. (2 point)

Plugging this tax rate back into the tax revenue formula $TR^* = TR(\tau_t^*)$ we see that the maximum tax revenue is $TR^* = \frac{\phi^\phi}{(1+\phi)^{(1+\phi)}}$. Therefore the maximum spending on defense is $G^* = \frac{\phi^\phi}{(1+\phi)^{(1+\phi)}}$. (1 point)

- (d) If the government DOES NOT run a balanced budget, the government can use debt to finance expenses. Suppose the government begins year t with debt B_t , collects income tax revenue using tax rate τ_t , spends G_t on defense. If the government faces an interest rate r on its debt what will be the debt next period B_{t+1} as a function of B_t, G_t, τ_t, r and ϕ ? (2 point)

Remember from class that for a given primary deficit in year t PD_t the evolution of government debt is $B_{t+1} = (1 + r)B_t + PD_t$. Since the primary deficit is $G_t - \tau_t(1 - \tau_t)^\phi$ the debt next period will be:

$$B_{t+1} = (1 + r)B_t + (G_t - \tau_t(1 - \tau_t)^\phi). \text{(2point)} \quad (8)$$

- (e) Now suppose that tax instruments are chosen so that the debt is constant across periods (i.e. $B_{t+1} = B_t = B$), taxes are constant across periods (i.e. $\tau_t = \tau$) and defense spending is constant across periods (i.e. $G_t = G$). Using the equation for the evolution of government debt from year to year, what will the sustainable debt level B equal as a function of τ, G, r , and ϕ ? (4 point)
Beginning with the equation of the evolution of debt from the previous problem we make all variables constant (i.e. independent of t) so that:

$$B = (1 + r)B + (G - \tau(1 - \tau)^\phi). \text{(2point)} \quad (9)$$

By grouping together terms we can solve for the sustainable debt level B :

$$B = \frac{1}{r}(\tau(1 - \tau)^\phi - G). \text{(2point)} \quad (10)$$

- (f) If the government runs a primary deficit each period and the interest rate is positive (so that $r > g$), can the government sustain a constant positive debt level B ? (2 point)

No it can't. In order for the government to sustain primary deficits and a constant positive debt level the interest rate must be less than the growth rate $r < g$. The solution for the constant sustainable debt level from the previous problem makes this clear. If the interest rate is positive $r > 0$ – thus greater than the growth rate $r > g$ – and the government runs a primary deficit $(\tau(1 - \tau)^\phi - G) < 0$ the sustainable debt level must be negative $B < 0$. (2 point)

Further explanation: Let see how this problem would work if there were growth in the economy (i.e. $g > 0$). This complicates the problem so let's walk through it together. Suppose there was growth in this economy such that output grew at a rate of g (so that $Y_{t+1} = (1 + g)Y_t$). Now lets divide the left and right-hand sides of the intertemporal government budget (from part d) by Y_{t+1} and $(1 + g)Y_t$ respectively:

$$\frac{B_{t+1}}{Y_{t+1}} = \frac{(1 + r)}{(1 + g)}\left(\frac{B_t}{Y_t}\right) + \left(\frac{G_t - \tau_t(1 - \tau_t)^\phi}{(1 + g)Y_t}\right). \quad (11)$$

Again make all variables constant, including output ($Y_{t+1} = Y_t = Y$):

$$\frac{B}{Y} = \frac{(1 + r)}{(1 + g)}\left(\frac{B}{Y}\right) + \left(\frac{G - \tau(1 - \tau)^\phi}{(1 + g)Y}\right). \quad (12)$$

Where $\frac{B}{Y}$ is the debt-to-GDP ratio and $\left(\frac{G - \tau(1 - \tau)^\phi}{Y}\right)$ is the deficit-to-GDP ratio. Now we solve for the sustainable debt-to-GDP ratio $\frac{B}{Y}$:

$$\frac{B}{Y} = \frac{1}{(r - g)}\left(\frac{\tau(1 - \tau)^\phi - G}{Y}\right). \quad (13)$$

Now we can see that if the interest rate is positive but less than the growth rate, $0 < r < g$, so that $r - g < 0$, and the government sustains a constant deficit-to-GDP ratio, $(\frac{\tau(1-\tau)^{\phi}-G}{Y}) < 0$, the sustainable debt-to-GDP ratio will be positive $\frac{B}{Y} > 0$. The problem on the exam is a specific case of this setup with $Y = 1$ and $g = 0$.

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